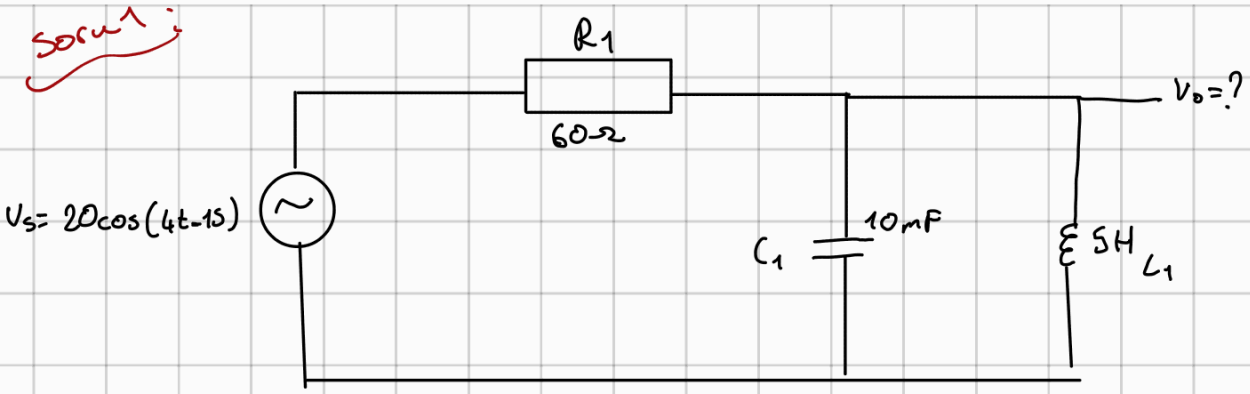
Soru 1 :




$v_s(t) = 20\cos(4t-15) \Rightarrow V_s = 20\, e^{j(-15)}$ $\quad \omega = 4$ rad/s

$R_1 = 60\Omega \Rightarrow Z_{R1} = 60\Omega$

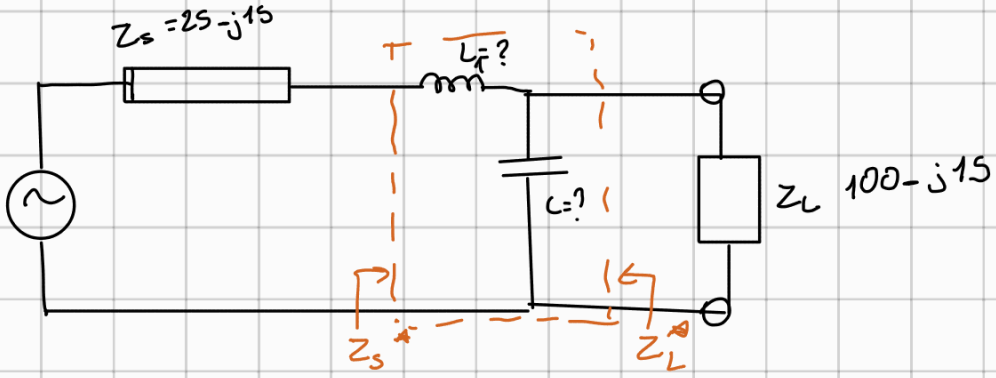
$C_1 = 10mF = 10.10^{-3}F \Rightarrow Z_{C1} = \frac{1}{j\omega C} = \frac{1}{j\,4.10^{-2}} = -j25$

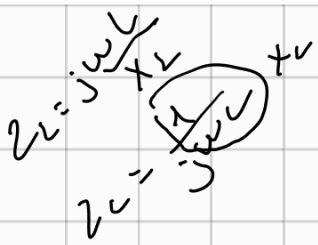
$L_1 = 5H \Rightarrow Z_{L1} = j\omega L = j\,4.5 = 20j$

$$\frac{1}{Z_{L1 \| C1}} = \frac{1}{Z_{L1}} + \frac{1}{Z_{C1}} = \frac{Z_{C1} + Z_{L1}}{Z_{C1}\, Z_{L1}} = \frac{-j25 + j20}{-j25 \cdot j20} = \frac{1}{100j}$$

$$V_o = V_s \cdot \frac{Z_{L1 \| C1}}{Z_{R1} + Z_{L1 \| C1}} = 20\angle -15 \cdot \frac{100j}{60 + 100j} \Rightarrow V_o = 17{,}15\angle 16^\circ$$

$v_o(t) = 17{,}15\cos($



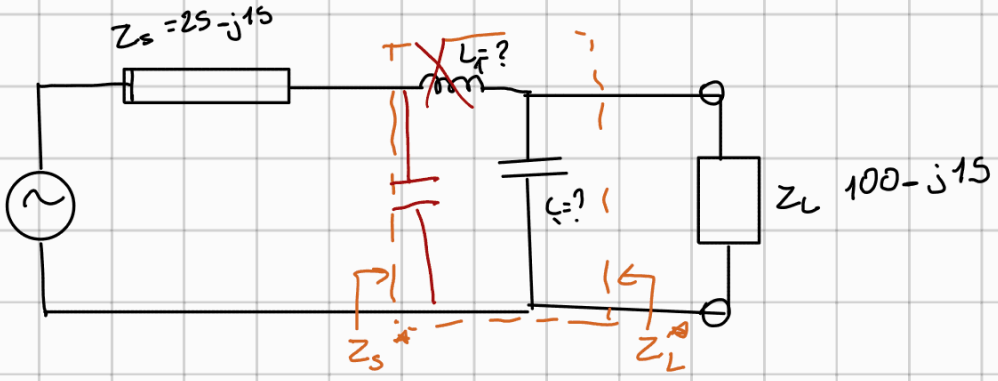



$Z_{L1} = j\omega L$

$Z_{L1} = j\omega L$ $\quad Z_C = \frac{1}{j\omega C} = -j\frac{1}{\omega C}$ $\quad Z_C = \frac{1}{j\omega L = C}$

$$25 + j15 = Z_L // Z_{C1} + Z_{L1}$$
$$25 + j15 = 100 - j15$$






$$Z_C = \frac{1}{j\omega C}$$

$$Z_L // Z_C = 25 + j15$$

$$\frac{100 - j15 + \frac{1}{j\omega C}}{\frac{100}{j\omega C} - \frac{j15}{j\omega C}} = 25 + j15$$

$$\frac{100 - j\left(15 + \frac{1}{\omega C}\right)}{-\frac{15}{\omega C} - j\frac{1}{\omega C}} = 25 + j15$$

$$\frac{100 - j\left(\frac{15\omega C + 1}{\omega C}\right)}{-(1+j)\frac{1}{\omega C}} = 25 + j15$$

$$\frac{100\omega C - j(15\omega C + 1)}{-(1+j)} = 25 + j15$$

$$100\omega C - j(15\omega C + 1) = -(25 + j15 + j25 - 15)$$

$$\Rightarrow 100\omega C - j(15\omega C + 1) = -10 - j40$$

$$100\omega C = -10 \qquad 15\omega C + 1 = 40$$

$$\omega C = \frac{-1}{10} \qquad 15\omega C = 39$$

$$\omega C = \frac{39}{15}$$